MSG No. 175.138 $P6/m1'$ [Type II, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `6/m..1'`

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: `1b`, site symmetry: `6/m..1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: `2c`, site symmetry: `-6..1'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,3,5,8,10,12,13,15,17,20,22,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[2,4,6,7,9,11,14,16,18,19,21,23]

Table 4: Wyckoff site: `2d`, site symmetry: `-6..1'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,3,5,8,10,12,13,15,17,20,22,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[2,4,6,7,9,11,14,16,18,19,21,23]

Table 5: Wyckoff site: `2e`, site symmetry: `6..1'`

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,4,5,6,13,14,15,16,17,18]
2	$[0, 0, -z]$	[7,8,9,10,11,12,19,20,21,22,23,24]

Table 6: Wyckoff site: `3f`, site symmetry: `2/m..1'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,7,10,13,16,19,22]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,5,8,11,14,17,20,23]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,6,9,12,15,18,21,24]

Table 7: Wyckoff site: `3g`, site symmetry: `2/m..1'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,4,7,10,13,16,19,22]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,5,8,11,14,17,20,23]

3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[3,6,9,12,15,18,21,24]

Table 8: Wyckoff site: 4h, site symmetry: 3..1'

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[1,3,5,13,15,17]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	[2,4,6,14,16,18]
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	[7,9,11,19,21,23]
4	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	[8,10,12,20,22,24]

Table 9: Wyckoff site: 6i, site symmetry: 2..1'

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	[1,4,13,16]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[2,5,14,17]
3	$\left[0,\ \frac{1}{2},\ z\right]$	[3,6,15,18]
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	[7,10,19,22]
5	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[8,11,20,23]
6	$\left[0,\ \frac{1}{2},\ -z\right]$	[9,12,21,24]

Table 10: Wyckoff site: 6j, site symmetry: m..1'

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,10,13,22]
2	$[x-y,\ x,\ 0]$	[2,11,14,23]
3	$[-y,\ x-y,\ 0]$	[3,12,15,24]
4	$[-x,\ -y,\ 0]$	[4,7,16,19]
5	$[-x+y,\ -x,\ 0]$	[5,8,17,20]
6	$[y,\ -x+y,\ 0]$	[6,9,18,21]

Table 11: Wyckoff site: 6k, site symmetry: m..1'

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	[1,10,13,22]
2	$\left[x-y,\ x,\ \frac{1}{2}\right]$	[2,11,14,23]
3	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[3,12,15,24]
4	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[4,7,16,19]
5	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[5,8,17,20]
6	$\left[y,\ -x+y,\ \frac{1}{2}\right]$	[6,9,18,21]

Table 12: Wyckoff site: 12l, site symmetry: 11'

No.	position	mapping
1	$[x,\ y,\ z]$	[1,13]
2	$[x-y,\ x,\ z]$	[2,14]
3	$[-y,\ x-y,\ z]$	[3,15]
4	$[-x,\ -y,\ z]$	[4,16]
5	$[-x+y,\ -x,\ z]$	[5,17]
6	$[y,\ -x+y,\ z]$	[6,18]
7	$[-x,\ -y,\ -z]$	[7,19]
8	$[-x+y,\ -x,\ -z]$	[8,20]
9	$[y,\ -x+y,\ -z]$	[9,21]
10	$[x,\ y,\ -z]$	[10,22]
11	$[x-y,\ x,\ -z]$	[11,23]
12	$[-y,\ x-y,\ -z]$	[12,24]